

Forcing Quasirandomness via Rooted F -Densities

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Abstract

We study rooted subgraph density equations in graphon-type limit objects. We resolve Lovász's two-rooted triangle problem by showing that the solutions of his triangle equation are precisely the triangle-free graphons and the constant graphons. More generally, we prove an all-edge rigidity theorem: for every finite graph F , if all edge-rooted F -densities in a graphon W are constant, then either $t(F, W) = 0$ or W is a constant graphon. Thus edge-transitive graphs require only one rooted edge.

The proof recasts rooted-density constancy as uniformity of edge marginals for the Gibbs measure associated with F . An entropy comparison and a Hoeffding variance decomposition then force $\log W$ to have no nonconstant component. The same argument gives quantitative stability for graphons bounded away from zero.

We extend the method to symmetric hyperkernels and full-edge-rooted Aldous–Hoover hypergraphons. For directed kernels, the exact obstruction is the Eulerian gauge $p \exp(\phi(x) - \phi(y))$, while non-Eulerian directed templates are rigid. In tournamentons this gauge is eliminated by the identity $W(x, y) + W(y, x) = 1$, so all-arc rooted constancy forces $W \equiv 1/2$, except in the zero-density case.

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1 Introduction

Graphons are the analytic limit objects for convergent dense graph sequences. A graphon is a symmetric measurable function $W : [0, 1]^2 \rightarrow [0, 1]$, considered up to measure-preserving rearrangement, and the homomorphism density of a finite graph $F = (V, E)$ is

$$t(F, W) = \int_{[0, 1]^V} \prod_{ij \in E} W(x_i, x_j) \prod_{i \in V} dx_i. \quad (1)$$

The graphon point of view was introduced by Lovász and Szegedy [11]; see also the monograph [7]. It gives a concise language for quasirandomness: many quasirandomness criteria say that a system of graphon equations has only constant solutions, paralleling the classical theorem of Chung, Graham and Wilson [3].

Lovász's 2008 problem list on graph homomorphisms introduced weakly homogeneous graphons in connection with finite forcibility and asked for a characterization of the first nontrivial case: graphons satisfying

$$W(x, y) \int_0^1 W(x, z) W(y, z) dz = \text{constant} \quad (2)$$

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for almost every (x, y) . This is Problem 28 in [8]. The surrounding section of the problem list asks for a characterization of weakly homogeneous and finitely forcible graphons; Lovasz and Szegedy later developed the finitely forcible graphon theory in [12].

The aim of this paper is to place (2) in a broader framework. If F is a finite graph and $e = ab \in E(F)$, define the edge-rooted density

$$r_e^W(x, y) = \int_{[0,1]^{V(F) \setminus \{a,b\}}} \prod_{ij \in E(F)} W(z_i, z_j) \prod_{i \in V(F) \setminus \{a,b\}} dz_i, \quad (3)$$

where $z_a = x$ and $z_b = y$. For $F = K_3$, this is exactly (2). Our main theorem says that the constancy of all edge-rooted densities of a fixed finite graph forces the graphon to be quasirandom unless the graph has zero density.

Theorem 1.1. *Let F be a finite graph with $m = |E(F)| \geq 1$, and let $W : [0, 1]^2 \rightarrow [0, 1]$ be a graphon. If r_e^W is constant for every $e \in E(F)$, then either $t(F, W) = 0$ or $W = p$ almost everywhere, where $p = t(F, W)^{1/m}$.*

If F is edge-transitive, one edge is enough. In particular, (2) has exactly two types of solutions: triangle-free graphons and constant graphons.

The common mechanism is developed in Section 2. Rooted-density constancy becomes uniformity of the corresponding marginals of a Gibbs measure. Entropy and Jensen's inequality then reduce the problem to a pair-sum rigidity statement for $\log W$, which is handled by Hoeffding decompositions.

The framework is robust: we prove quantitative stability versions showing that approximate marginal uniformity forces $\log W$ to be close in L^2 to a constant, with analogous estimates for symmetric k -uniform hyperkernels, full Aldous-Hoover hypergraphons, and non-Eulerian directed kernels.

Theorem 1.2. *Let H be a finite k -uniform hypergraph with $m = |E(H)| \geq 1$.*

- (i) *If $K : [0, 1]^k \rightarrow [0, 1]$ is a symmetric k -uniform hyperkernel and r_e^K is constant for every $e \in E(H)$, then either $t(H, K) = 0$ or $K = p$ almost everywhere, where $p = t(H, K)^{1/m}$.*
- (ii) *In the Aldous-Hoover hypergraphon representation, if every full-edge-rooted density of H in a k -uniform hypergraphon W is constant, then either $t(H, W) = 0$ or W is constant almost everywhere.*

Hypergraph limits require additional coordinates beyond vertex coordinates; this is why the correct rooted object in the hypergraphon setting is a full edge, not merely the vertices of an edge. Hypergraph limit objects in this form were constructed by Elek and Szegedy [5] and by Zhao [15]; Zhao's construction explicitly uses functions on $[0, 1]^{2^k - 2}$.

Directed kernels exhibit a new phenomenon. Let D be a finite oriented graph and $W : [0, 1]^2 \rightarrow [0, 1]$ an asymmetric kernel. The same entropy reduction leads to a directed pair-sum problem.

Theorem 1.3. *Let $D = (V, A)$ be a finite oriented graph with $m = |A| \geq 1$, and let $W : [0, 1]^2 \rightarrow [0, 1]$ be a directed kernel. Suppose r_a^W is constant for every arc $a \in A$ and $t(D, W) > 0$.*

- (i) *If D is not Eulerian, then W is constant almost everywhere.*
- (ii) *If D is Eulerian, then*

$$W(x, y) = p \exp(\phi(x) - \phi(y))$$

for some $p \in (0, 1]$ and bounded measurable ϕ .

Conversely, every kernel of this form with $0 \leq W \leq 1$ has all arc-rooted densities of every Eulerian D equal to the constant p^m .

Thus the naive directed analogue of Theorem 1.1 is false.

Tournament limits form a constrained subclass of the directed theory in which rigidity is restored. A tournamenton is a directed kernel $W : [0, 1]^2 \rightarrow [0, 1]$ satisfying

$$W(x, y) + W(y, x) = 1$$

for almost every (x, y) . Tournament limits were developed, for instance, by Thornblad [14]. We prove that this extra identity eliminates the directed gauge obstruction.

Theorem 1.4. *Let T be a finite tournament with at least one arc, and let W be a tournamenton. If the arc-rooted density r_a^W is constant for every arc $a \in A(T)$, then either $t(T, W) = 0$ or $W = 1/2$ almost everywhere. If T is arc-transitive, the same conclusion follows from the constancy of r_a^W for a single arc a .*

2 Preliminaries

2.1 Rooted densities and Gibbs marginals

All graphons are represented by measurable symmetric functions $W : [0, 1]^2 \rightarrow [0, 1]$. If $F = (V, E)$ is a finite graph, $t(F, W)$ is defined by (1). For $e = ab \in E$, the rooted density r_e^W is defined by (3). It satisfies

$$\int_{[0,1]^2} r_e^W(x, y) dx dy = t(F, W). \quad (4)$$

Hence, if r_e^W is constant, then it is identically $t(F, W)$.

When $Z = t(F, W) > 0$, define the F -Gibbs measure in W by

$$\frac{d\mu_{F,W}}{d\lambda^V}(\mathbf{x}) = \frac{1}{Z} \prod_{ij \in E} W(x_i, x_j). \quad (5)$$

The marginal density of (x_a, x_b) under $\mu_{F,W}$ is

$$q_e^W(x, y) = \frac{r_e^W(x, y)}{Z}. \quad (6)$$

Thus r_e^W is constant if and only if $q_e^W \equiv 1$.

2.2 The entropy reduction

We use relative entropy in the following form. If μ has density ρ with respect to a probability measure ν , then

$$D(\mu \| \nu) = \int \rho \log \rho d\nu \geq 0,$$

with equality if and only if $\rho = 1$ almost everywhere.

Proposition 2.1 (Entropy reduction for graphons). *Let $F = (V, E)$ be a finite graph with $m = |E| \geq 1$, let W be a graphon, and put $Z = t(F, W)$. Suppose $Z > 0$ and $q_e^W \equiv 1$ for every $e \in E$. Then $u = \log W$ is bounded and*

$$\sum_{ij \in E} u(x_i, x_j) = \log Z \quad (7)$$

for almost every $\mathbf{x} \in [0, 1]^V$.

Proof. Since $q_e^W \equiv 1$, we have $r_e^W \equiv Z$ for every edge e . Fixing $e = ab$ and using that all other factors in the rooted density are at most 1 gives

$$Z = r_e^W(x, y) \leq W(x, y)$$

for almost every (x, y) . Hence $W \geq Z > 0$, and $u = \log W$ is bounded.

Let $\mu = \mu_{F,W}$ be the Gibbs measure (5). Since each edge marginal is uniform,

$$D(\mu \| \lambda^V) = \sum_{e \in E} \int_{[0,1]^2} u(x, y) q_e^W(x, y) dx dy - \log Z = m\bar{u} - \log Z \geq 0,$$

where $\bar{u} = \int_{[0,1]^2} u$. Thus $m\bar{u} \geq \log Z$. On the other hand, Jensen's inequality gives

$$\log Z = \log \int \exp \left(\sum_{ij \in E} u(x_i, x_j) \right) \prod_i dx_i \geq m\bar{u}.$$

Equality holds. Since the exponential function is strictly convex, (7) follows. $\square$

2.3 Hoeffding decomposition and pair-sum rigidity

Lemma 2.2 (Hoeffding decomposition for symmetric kernels). *Let $u \in L^2([0, 1]^2)$ be symmetric. There are unique functions $f \in L^2([0, 1])$ and $g \in L^2([0, 1]^2)$ such that*

$$u(x, y) = \bar{u} + f(x) + f(y) + g(x, y), \quad (8)$$

where

$$\bar{u} = \int_{[0, 1]^2} u(x, y) dx dy,$$

$\int f = 0$, g is symmetric, and $\int_0^1 g(x, y) dy = 0$ for almost every x . Moreover,

$$\|u - \bar{u}\|_2^2 = 2\|f\|_2^2 + \|g\|_2^2. \quad (9)$$

Proof. Set

$$f(x) = \int_0^1 u(x, y) dy - \bar{u}, \quad g(x, y) = u(x, y) - \bar{u} - f(x) - f(y).$$

The stated mean-zero and degeneracy properties are immediate. Orthogonality of the three nonconstant components follows from Fubini. For example,

$$\int f(x)g(x, y) dx dy = \int f(x) \left(\int g(x, y) dy \right) dx = 0,$$

and $\int f(x)f(y) dx dy = 0$. This proves (9). Uniqueness follows by taking the two-variable mean and one-variable marginals. $\square$

Lemma 2.3 (Variance identity). *Let $F = (V, E)$ have $m = |E| \geq 1$. Let $u \in L^2([0, 1]^2)$ be symmetric, and let*

$$S(\mathbf{X}) = \sum_{ij \in E} u(X_i, X_j),$$

where $(X_i)_{i \in V}$ are independent uniform random variables. Then

$$\|u - \bar{u}\|_2^2 \leq \frac{1}{m} \text{Var}(S). \quad (10)$$

Proof. Use (8). Then

$$S - m\bar{u} = \sum_{v \in V} d_F(v)f(X_v) + \sum_{ij \in E} g(X_i, X_j).$$

The two sums are orthogonal. The variables $f(X_v)$ are orthogonal for different vertices, and the variables $g(X_i, X_j)$ are orthogonal for different edges: if two edges share a vertex, degeneracy of g gives zero covariance; if they are disjoint, independence gives zero covariance. Hence

$$\text{Var}(S) = \left(\sum_{v \in V} d_F(v)^2 \right) \|f\|_2^2 + m \|g\|_2^2.$$

Since $\sum_v d_F(v)^2 \geq \sum_v d_F(v) = 2m$, this and (9) imply (10). $\square$

Lemma 2.4 (Pair-sum rigidity). *If $u \in L^2([0, 1]^2)$ is symmetric and*

$$\sum_{ij \in E(F)} u(x_i, x_j) = C$$

for almost every $\mathbf{x} \in [0, 1]^{V(F)}$, where $E(F) \neq \emptyset$, then u is constant almost everywhere.

Proof. The hypothesis says that the variance of the sum is zero. Apply Lemma 2.3. $\square$

2.4 An exponential deficit estimate

Lemma 2.5 (Exponential deficit controls variance). *Let X be a real random variable with $\mathbb{E}X = 0$ and $|X| \leq R$ almost surely. Then*

$$\text{Var}(X) \leq (2e^R + R^2) \log \mathbb{E}e^X. \quad (11)$$

Proof. For $|x| \leq R$,

$$e^x \geq 1 + x + \frac{e^{-R}}{2} x^2.$$

Taking expectations gives $\mathbb{E}e^X \geq 1 + e^{-R} \text{Var}(X)/2$. Since $\text{Var}(X) \leq R^2$, the inequality $\log(1+s) \geq s/(1+s)$ yields

$$\log \mathbb{E}e^X \geq \frac{e^{-R} \text{Var}(X)/2}{1 + e^{-R} R^2/2}.$$

Rearranging proves the claim. $\square$

3 The graphon theorem and stability

3.1 Exact rigidity

Theorem 3.1 (Edge-marginal rigidity). *Let F be a finite graph with $m = |E(F)| \geq 1$, and let W be a graphon. If r_e^W is constant for every edge $e \in E(F)$, then either $t(F, W) = 0$ or $W = p$ almost everywhere, where $p = t(F, W)^{1/m}$.*

Proof. Let $Z = t(F, W)$. If $Z = 0$, there is nothing to prove. Assume $Z > 0$. By (4), each constant rooted density is equal to Z . Thus $q_e^W \equiv 1$ for every $e \in E$.

By Proposition 2.1, $u = \log W$ satisfies

$$\sum_{ij \in E} u(x_i, x_j) = \log Z$$

for almost every $\mathbf{x}$. By Lemma 2.4, u is constant. Hence W is constant, say $W = p$. Then $Z = p^m$, so $p = Z^{1/m}$. $\square$

Corollary 3.2 (Edge-transitive graphs). *Let F be edge-transitive and let $e \in E(F)$. If r_e^W is constant, then either $t(F, W) = 0$ or W is constant.*

Proof. Every edge of F is the image of e under an automorphism. Rooted densities are therefore identical, up to a possible coordinate interchange, and W is symmetric. Hence all edge-rooted densities are constant, and Theorem 3.1 applies. $\square$

Corollary 3.3 (Lovasz's two-rooted triangle problem). *For a graphon W , the function*

$$W(x, y) \int_0^1 W(x, z) W(y, z) dz$$

is constant if and only if either $t(K_3, W) = 0$ or W is constant. If the constant value is $c > 0$, then $W = c^{1/3}$ almost everywhere.

Proof. This is Corollary 3.2 applied to K_3 . If $t(K_3, W) = 0$, then

$$W(x, y) W(x, z) W(y, z) = 0$$

almost everywhere, which is precisely triangle-freeness. If $W = p$, the rooted density is p^3 . $\square$

3.2 Quantitative stability

Theorem 3.4 (Graphon stability). *Let F be a finite graph with $m = |E(F)| \geq 1$. Suppose $a \leq W \leq 1$ almost everywhere, where $a \in (0, 1]$, and put $L = \log(1/a)$. Define*

$$\Delta_F(W) = \sum_{e \in E(F)} \|q_e^W - 1\|_{L^1([0,1]^2)}.$$

Then, with $\bar{u} = \int \log W$,

$$\|\log W - \bar{u}\|_2^2 \leq \frac{1}{m}(2e^{mL} + m^2 L^2) L \Delta_F(W). \quad (12)$$

Consequently, if $p = e^{\bar{u}}$, then

$$\|W - p\|_2^2 \leq \frac{1}{m}(2e^{mL} + m^2 L^2) L \Delta_F(W),$$

and the same upper bound controls $\|W - p\|_1^2$.

Proof. Let $u = \log W$, $Z = t(F, W)$, and $S = \sum_{ij \in E} u(X_i, X_j)$ for independent uniform X_i . Let

$$J = \log Z - m\bar{u} \geq 0$$

be the Jensen deficit. The Gibbs entropy identity gives

$$D(\mu_{F,W} \|\lambda^V) = -J + \sum_{e \in E} \int u(x, y) (q_e^W(x, y) - 1) dx dy.$$

Since entropy is nonnegative and $|u| \leq L$,

$$J \leq L \Delta_F(W). \quad (13)$$

Now $X = S - \mathbb{E}S$ satisfies $|X| \leq mL$ and

$$\log \mathbb{E}e^X = J.$$

By [Lemma 2.5](#),

$$\text{Var}(S) \leq (2e^{mL} + m^2 L^2) J.$$

Use [Lemma 2.3](#) and (13) to obtain (12). Finally, $x \mapsto e^x$ is 1-Lipschitz on $[-L, 0]$, so $\|W - p\|_2 \leq \|u - \bar{u}\|_2$; and $\|W - p\|_1 \leq \|W - p\|_2$. $\square$

4 Hyperkernels and hypergraphons

We shall use the following finite-product form of the Hoeffding, or ANOVA, decomposition.

Lemma 4.1 (Hoeffding decomposition on a finite product). *Let I be a finite set and let $u \in L^2([0, 1]^I)$. There are unique functions $u_R \in L^2([0, 1]^R)$, $R \subseteq I$, such that*

$$u(x_I) = \sum_{R \subseteq I} u_R(x_R),$$

where $u_\emptyset = \int u$ and, for every nonempty R and every $i \in R$,

$$\int_0^1 u_R(x_R) dx_i = 0$$

for almost every choice of the remaining variables in R . The functions u_R are mutually orthogonal after being viewed as functions on $[0, 1]^I$, and therefore

$$\|u\|_2^2 = \sum_{R \subseteq I} \|u_R\|_2^2.$$

Equivalently,

$$\left\| u - \int u \right\|_2^2 = \sum_{\emptyset \neq R \subseteq I} \|u_R\|_2^2.$$

If a finite group Γ acts on I and u is invariant under this action, then the decomposition is equivariant: for every $\gamma \in \Gamma$ the component indexed by γR is the relabelled copy of the component indexed by R .

Proof. For $R \subseteq I$, let $E_R u = \mathbb{E}[u \mid x_R]$, with $E_\emptyset u = \int u$. Define

$$u_R = \sum_{S \subseteq R} (-1)^{|R|-|S|} E_S u.$$

Möbius inversion on the subset lattice gives $u = \sum_{R \subseteq I} u_R$. If $i \in R$, then integrating u_R in the coordinate i replaces every $E_S u$ with $E_{S \setminus \{i\}} u$; the terms cancel in pairs, proving degeneracy. If $R \neq T$, choose $i \in R \setminus T$ or $i \in T \setminus R$; integrating first in that coordinate shows u_R and u_T are orthogonal. The norm identity follows. Uniqueness follows by applying the conditional expectation operators and Möbius inversion again. Finally, if u is invariant under a group action, the relabelled family of components satisfies the same defining properties as the original family, so uniqueness gives equivariance. $\square$

4.1 Symmetric uniform hyperkernels

Let $k \geq 2$. A symmetric k -uniform hyperkernel is a measurable function

$$K : [0, 1]^k \rightarrow [0, 1]$$

that is invariant under permutations of the coordinates. If $H = (V, E)$ is a finite k -uniform hypergraph, define

$$t(H, K) = \int_{[0, 1]^V} \prod_{e \in E} K(x_e) \prod_{v \in V} dx_v, \quad (14)$$

where x_e denotes the k -tuple $(x_v)_{v \in e}$ in any order. For $e \in E$, the edge-rooted density is

$$r_e^K(\mathbf{x}_e) = \int \prod_{f \in E} K(\mathbf{z}_f) \prod_{v \notin e} dz_v, \quad (15)$$

where the variables on e are fixed to be $\mathbf{x}_e$.

Lemma 4.2 (Symmetric k -coordinate Hoeffding decomposition). *Let $u \in L^2([0, 1]^k)$ be invariant under all permutations of its coordinates. There are unique symmetric kernels $u_s \in L^2([0, 1]^s)$, $1 \leq s \leq k$, degenerate in each coordinate, such that*

$$u(x_1, \dots, x_k) = \bar{u} + \sum_{s=1}^k \sum_{A \in \binom{[k]}{s}} u_s(x_A). \quad (16)$$

The summands in (16) are pairwise orthogonal after they are viewed as functions on $[0, 1]^k$, and

$$\|u - \bar{u}\|_2^2 = \sum_{s=1}^k \binom{k}{s} \|u_s\|_2^2. \quad (17)$$

Proof. Apply Lemma 4.1 with $I = [k]$. If two coordinate sets $A, A' \subseteq [k]$ have the same size, a permutation sends A to A' . Equivariance and the symmetry of u imply that the corresponding Hoeffding components are the same function after relabelling coordinates. This common component is denoted by u_s , where $s = |A|$. Orthogonality and (17) are the orthogonality and norm identity in Lemma 4.1. $\square$

Lemma 4.3 (Hyperkernel variance lower bound). *Let $H = (V, E)$ be a finite k -uniform hypergraph with $m = |E| \geq 1$. If $u \in L^2([0, 1]^k)$ is symmetric and*

$$S_H(u) = \sum_{e \in E} u(X_e),$$

where $(X_v)_{v \in V}$ are independent uniform variables, then

$$\text{Var}(S_H(u)) \geq m \|u - \bar{u}\|_2^2. \quad (18)$$

Consequently, if $S_H(u)$ is almost surely constant, then u is constant almost everywhere.

Proof. Use Lemma 4.2. For $B \subseteq V$ with $|B| = s$, let

$$c_B = \#\{e \in E : B \subseteq e\}.$$

Then

$$S_H(u) - m\bar{u} = \sum_{s=1}^k \sum_{B \in \binom{V}{s}} c_B u_s(X_B).$$

The terms indexed by distinct vertex sets B are orthogonal by degeneracy, exactly as in Lemma 4.1. Therefore

$$\text{Var}(S_H(u)) = \sum_{s=1}^k \sum_{B \in \binom{V}{s}} c_B^2 \|u_s\|_2^2.$$

For each fixed s , the integers c_B satisfy

$$\sum_{B \in \binom{V}{s}} c_B = m \binom{k}{s}, \quad \sum_{B \in \binom{V}{s}} c_B^2 \geq \sum_{B \in \binom{V}{s}} c_B.$$

Combining this with (17) gives (18). If $S_H(u)$ is constant, then the variance is zero and hence $u = \bar{u}$ in L^2 . $\square$

Theorem 4.4 (Hyperkernel edge-marginal rigidity). *Let H be a finite k -uniform hypergraph with $m = |E(H)| \geq 1$. If $K : [0, 1]^k \rightarrow [0, 1]$ is a symmetric k -uniform hyperkernel and r_e^K is constant for every $e \in E(H)$, then either $t(H, K) = 0$ or $K = p$ almost everywhere, where $p = t(H, K)^{1/m}$.*

Proof. Let $Z = t(H, K)$. If $Z = 0$, there is nothing to prove, so assume $Z > 0$. Integrating (15) over the rooted variables gives $t(H, K)$, so the constant value of each r_e^K is Z . Thus $r_e^K \equiv Z$ for every $e \in E(H)$. Since all factors in (15) except the rooted factor are at most 1, we have $K \geq Z$ almost everywhere. Thus $u = \log K$ is bounded.

Let μ be the H -Gibbs measure with density $Z^{-1} \prod_{e \in E(H)} K(x_e)$. The rooted hypotheses say that every k -dimensional edge marginal of μ is Lebesgue measure. Therefore

$$D(\mu \| \lambda^V) = m \int_{[0, 1]^k} u - \log Z \geq 0.$$

Jensen's inequality gives

$$\log Z = \log \mathbb{E} \exp \left(\sum_{e \in E(H)} u(X_e) \right) \geq m \int_{[0, 1]^k} u.$$

Equality holds, and the strict convexity of the exponential implies

$$\sum_{e \in E(H)} u(X_e) = \log Z$$

almost surely. By Lemma 4.3, u is constant. Hence $K = p$ almost everywhere and $p = Z^{1/m}$. $\square$

Theorem 4.5 (Hyperkernel stability). *Let H be a finite k -uniform hypergraph with $m \geq 1$ edges. Suppose $a \leq K \leq 1$ and put $L = \log(1/a)$. Define normalized edge marginals $q_e^K = r_e^K / t(H, K)$ and*

$$\Delta_H(K) = \sum_{e \in E(H)} \|q_e^K - 1\|_{L^1([0,1]^k)}.$$

Then

$$\left\| \log K - \int \log K \right\|_2^2 \leq \frac{1}{m} (2e^{mL} + m^2 L^2) L \Delta_H(K). \quad (19)$$

Consequently K is close in L^2 and L^1 to the constant $\exp(\int \log K)$ with the same bound.

Proof. Let $u = \log K$, $Z = t(H, K)$, $\bar{u} = \int u$, and

$$S = \sum_{e \in E(H)} u(X_e).$$

The entropy calculation used in Theorem 3.4 gives the Jensen deficit bound

$$J := \log Z - m\bar{u} \leq L \Delta_H(K).$$

For $X = S - \mathbb{E}S$, we have $|X| \leq mL$ and $\log \mathbb{E}e^X = J$. Hence Lemma 2.5 gives

$$\text{Var}(S) \leq (2e^{mL} + m^2 L^2) L \Delta_H(K).$$

The lower bound Lemma 4.3 gives

$$m \|u - \bar{u}\|_2^2 \leq \text{Var}(S),$$

which proves (19). The final L^2 and L^1 conclusions follow from the same Lipschitz argument as in Theorem 3.4. $\square$

4.2 Full hypergraphons

The preceding theorem treats the naive symmetric kernel model $K : [0, 1]^k \rightarrow [0, 1]$. Genuine dense k -uniform hypergraph limits are richer. In one standard representation, a k -uniform hypergraphon is a symmetric measurable function

$$W : [0, 1]^{\mathcal{A}_k} \rightarrow [0, 1], \quad \mathcal{A}_k = \{A : \emptyset \neq A \subsetneq [k]\},$$

so $|\mathcal{A}_k| = 2^k - 2$. For a finite k -uniform hypergraph $H = (V, E)$, choose independent variables X_B indexed by all nonempty $B \subseteq V$ with $|B| < k$. For an edge $e \in E$, the argument X_e° of W is the tuple of all X_B with $\emptyset \neq B \subsetneq e$. Then

$$t(H, W) = \mathbb{E} \prod_{e \in E(H)} W(X_e^\circ). \quad (20)$$

A full-edge-rooted density fixes all coordinates X_B with $\emptyset \neq B \subsetneq e$, not merely the vertex variables $X_{\{v\}}$.

Lemma 4.6 (Full-edge Hoeffding variance lower bound). *Let $H = (V, E)$ be a nonempty k -uniform hypergraph and let $u \in L^2([0, 1]^{\mathcal{A}_k})$ be invariant under the natural action of the symmetric group S_k on $\mathcal{A}_k$. Choose, for each edge $e \in E$, an ordering $\eta_e : [k] \rightarrow e$, and write*

$$S_H(u) = \sum_{e \in E} u(X_e^\circ).$$

Then

$$\text{Var}(S_H(u)) \geq |E| \|u - \bar{u}\|_2^2. \quad (21)$$

In particular, if $S_H(u)$ is almost surely constant, then u is constant almost everywhere.

Proof. Apply [Lemma 4.1](#) to the local coordinate set $\mathcal{A}_k$:

$$u = \bar{u} + \sum_{\emptyset \neq R \subseteq \mathcal{A}_k} u_R, \quad (22)$$

where u_R depends only on the local coordinates in R and is degenerate in each of them. For an edge e and $A \in \mathcal{A}_k$, put

$$\eta_e(A) = \{\eta_e(i) : i \in A\}.$$

For a nonempty $R \subseteq \mathcal{A}_k$, write

$$\eta_e(R) = \{\eta_e(A) : A \in R\}.$$

This is a family of global coordinates. Denote by

$$u_{e,R} := u_R((X_{\eta_e(A)})_{A \in R})$$

the corresponding lifted function. Then

$$S_H(u) - |E|\bar{u} = \sum_{e \in E} \sum_{\emptyset \neq R \subseteq \mathcal{A}_k} u_{e,R}. \quad (23)$$

Each $u_{e,R}$ is a Hoeffding-degenerate function of precisely the global coordinates in the family $\eta_e(R)$. Therefore, applying [Lemma 4.1](#) to the full independent coordinate system

$$\{X_B : \emptyset \neq B \subseteq V, |B| < k\},$$

shows that $u_{e,R}$ and $u_{f,R'}$ are orthogonal whenever $\eta_e(R) \neq \eta_f(R')$.

We next prove that no cancellation can occur among the terms with the same global coordinate family. Suppose

$$\eta_e(R) = \eta_f(R') =: G.$$

Set

$$I_R = \bigcup_{A \in R} A, \quad I_{R'} = \bigcup_{A' \in R'} A'.$$

Taking unions in the equality $\eta_e(R) = \eta_f(R')$ gives

$$\eta_e(I_R) = \eta_f(I_{R'}).$$

Thus

$$\theta := \eta_f^{-1} \circ \eta_e : I_R \rightarrow I_{R'}$$

is a well-defined bijection. If $A \in R$, then $\eta_e(A) \in G$, so there is a unique $A' \in R'$ with $\eta_f(A') = \eta_e(A)$, and by definition this A' is $\theta(A)$. Hence $R' = \theta R$. Extend θ arbitrarily to a permutation $\pi \in S_k$; then $\pi R = R'$ and

$$\eta_f(\pi A) = \eta_e(A) \quad (A \in R). \quad (24)$$

By the equivariance part of [Lemma 4.1](#), the component $u_{\pi R}$ is the relabelled copy of u_R , namely

$$u_{\pi R}((z_{\pi A})_{A \in R}) = u_R((z_A)_{A \in R}) \quad (25)$$

for almost every local coordinate tuple. Since $R' = \pi R$ and $u_{R'} = u_{\pi R}$, equations (24) and (25) give

$$u_{f,R'} = u_{R'}((X_{\eta_f(A')})_{A' \in R'}) = u_R((X_{\eta_e(A)})_{A \in R}) = u_{e,R}.$$

Thus, inside a fixed global coordinate family G , all summands in (23) are the same function. Write this common function as v_G , and let n_G be its multiplicity.

The global Hoeffding subspaces indexed by distinct nonempty coordinate families G are mutually orthogonal. Therefore

$$\begin{aligned}\text{Var}(S_H(u)) &= \sum_G n_G^2 \|v_G\|_2^2 \\ &\geq \sum_G n_G \|v_G\|_2^2 \\ &= \sum_{e \in E} \sum_{\emptyset \neq R \subseteq \mathcal{A}_k} \|u_R\|_2^2 \\ &= |E| \|u - \bar{u}\|_2^2,\end{aligned}$$

where the last equality follows from the local orthogonal decomposition (22). If $S_H(u)$ is constant, then the variance is zero, and the displayed lower bound forces $u = \bar{u}$ almost everywhere. $\square$

Theorem 4.7 (Full-edge hypergraphon rigidity and stability). *Let H be a finite k -uniform hypergraph with $m = |E(H)| \geq 1$, and let $W : [0, 1]^{\mathcal{A}_k} \rightarrow [0, 1]$ be a k -uniform hypergraphon in the representation above. If every full-edge-rooted density is constant, then either $t(H, W) = 0$ or W is constant almost everywhere.*

Moreover, if $a \leq W \leq 1$ almost everywhere and $L = \log(1/a)$, define q_e^W to be the normalized full-edge marginal density and put

$$\Delta_H(W) = \sum_{e \in E(H)} \|q_e^W - 1\|_{L^1([0, 1]^{\mathcal{A}_k})}.$$

Then

$$\left\| \log W - \int \log W \right\|_2^2 \leq \frac{1}{m} (2e^{mL} + m^2 L^2) L \Delta_H(W).$$

Proof. We first prove the exact assertion. Let $Z = t(H, W)$. If $Z = 0$, there is nothing to prove, so assume $Z > 0$. Constancy of the full-edge-rooted density gives $r_e^W \equiv Z$ for every $e \in E(H)$. Since the other factors in the rooted integral are at most 1, we have $W \geq Z$ almost everywhere on its full local coordinate space. Hence $u = \log W$ is bounded.

The H -Gibbs measure has density $Z^{-1} \prod_{e \in E(H)} W(X_e^\circ)$. The full-edge-rooted hypotheses say that every full-edge marginal is the Lebesgue measure on $[0, 1]^{\mathcal{A}_k}$. Therefore

$$D(\mu \| \lambda) = m \int u - \log Z \geq 0,$$

while Jensen gives

$$\log Z = \log \mathbb{E} \exp \left(\sum_{e \in E(H)} u(X_e^\circ) \right) \geq m \int u.$$

Equality holds, and the equality case of Jensen yields

$$\sum_{e \in E(H)} u(X_e^\circ) = \log Z$$

almost surely. By Lemma 4.6, this forces u to be constant, and hence W is constant.

For the stability assertion, assume $a \leq W \leq 1$. Then $Z = t(H, W) > 0$. Put $u = \log W$, $\bar{u} = \int u$, and

$$S = \sum_{e \in E(H)} u(X_e^\circ).$$

The full-edge Gibbs entropy identity gives, exactly as in the graphon proof,

$$D(\mu \| \lambda) = -J + \sum_{e \in E(H)} \int u(q_e^W - 1), \quad J := \log Z - m\bar{u}.$$

Since entropy is nonnegative and $|u| \leq L$, we have

$$J \leq L\Delta_H(W).$$

For $X = S - \mathbb{E}S$, we have $|X| \leq mL$ and $\log \mathbb{E}e^X = J$. Hence [Lemma 2.5](#) gives

$$\text{Var}(S) \leq (2e^{mL} + m^2L^2)L\Delta_H(W).$$

Finally, [Lemma 4.6](#) gives

$$m \|u - \bar{u}\|_2^2 \leq \text{Var}(S),$$

and the stated stability bound follows. $\square$

Remark 4.8 (Why full-edge rooting is necessary). In the genuine hypergraphon model, rooting only the vertices of a hyperedge is insufficient. For $k = 3$, let

$$W = p + \varepsilon(f(x_{12}) + f(x_{13}) + f(x_{23})),$$

where $\int f = 0$ and ε is small enough that $0 \leq W \leq 1$. If one fixes only the three vertex variables and integrates the pair variables, the rooted density of a single hyperedge is constant, although W is not constant. Full-edge rooting sees the pair coordinates and removes this hidden structure.

5 Directed kernels and the gauge obstruction

Let $D = (V, A)$ be a finite oriented graph: there are no loops, no repeated arcs, and no pair of opposite arcs. A directed kernel is a measurable function $W : [0, 1]^2 \rightarrow [0, 1]$, not assumed symmetric. Define

$$t(D, W) = \int_{[0, 1]^V} \prod_{(i, j) \in A} W(x_i, x_j) \prod_{i \in V} dx_i. \quad (26)$$

For an arc $a = (i, j)$, define $r_a^W(x, y)$ by fixing $x_i = x$, $x_j = y$, and integrating all remaining vertex variables. When $t(D, W) > 0$, define the normalized arc marginal

$$q_a^W(x, y) = \frac{r_a^W(x, y)}{t(D, W)}.$$

5.1 The directed pair-sum lemma

Lemma 5.1 (Directed Hoeffding decomposition). *Every $u \in L^2([0, 1]^2)$ has a unique decomposition*

$$u(x, y) = \bar{u} + f(x) + g(y) + h(x, y), \quad (27)$$

where $\int f = \int g = 0$, $\int h(x, y) dy = 0$ for a.e. x , and $\int h(x, y) dx = 0$ for a.e. y . Moreover,

$$\|u - \bar{u}\|_2^2 = \|f\|_2^2 + \|g\|_2^2 + \|h\|_2^2.$$

Proof. Set

$$f(x) = \int u(x, y) dy - \bar{u}, \quad g(y) = \int u(x, y) dx - \bar{u},$$

then define $h = u - \bar{u} - f - g$. The definitions give $\int f = \int g = 0$ and the required row and column degeneracy of h .

The decomposition is unique. Indeed, taking the total mean of (27) determines $\bar{u}$. Taking conditional expectation with respect to the first coordinate gives

$$\mathbb{E}[u(x, Y)] - \bar{u} = f(x),$$

because $\int g = 0$ and $\int h(x, y) dy = 0$. Similarly, conditional expectation with respect to the second coordinate determines $g(y) = \mathbb{E}[u(X, y)] - \bar{u}$. Then $h = u - \bar{u} - f - g$ is also determined.

Orthogonality follows from the row and column degeneracy of h and the zero means of f and g : the functions $f(x)$, $g(y)$, and $h(x, y)$ are pairwise orthogonal in $L^2([0, 1]^2)$. This gives the stated norm identity. $\square$

For a directed graph D , write $d^+(v)$ and $d^-(v)$ for the outdegree and indegree of v . Define the 2×2 matrix

$$M_D = \begin{pmatrix} \sum_v d^+(v)^2 & \sum_v d^+(v)d^-(v) \\ \sum_v d^+(v)d^-(v) & \sum_v d^-(v)^2 \end{pmatrix}. \quad (28)$$

Let γ_D be its smallest eigenvalue. Equivalently,

$$\gamma_D = \inf_{(\alpha, \beta) \neq (0,0)} \frac{\sum_{v \in V} (d^+(v)\alpha + d^-(v)\beta)^2}{\alpha^2 + \beta^2}. \quad (29)$$

If D is non-Eulerian, then $\gamma_D > 0$: otherwise the degree vectors $(d^+(v), d^-(v))$ would all be collinear, and since $\sum_v d^+(v) = \sum_v d^-(v) = m$, their common line would be spanned by $(1, 1)$, forcing $d^+(v) = d^-(v)$ for every non-isolated vertex.

Lemma 5.2 (Directed pair-sum structure). *Let $D = (V, A)$ be a finite oriented graph with $m = |A| \geq 1$, and let $u \in L^2([0, 1]^2)$. If*

$$\sum_{(i,j) \in A} u(x_i, x_j) = C \quad (30)$$

for almost every $\mathbf{x} \in [0, 1]^V$, then:

- (i) if D is not Eulerian, then u is constant almost everywhere;
- (ii) if D is Eulerian, then $u(x, y) = c + \phi(x) - \phi(y)$ for some $c \in \mathbb{R}$ and $\phi \in L^2([0, 1])$.

Here D is Eulerian means $d^+(v) = d^-(v)$ for every non-isolated vertex v .

Proof. Use (27). Let X_v be independent uniform variables. The variance of the left side of (30) equals the variance of

$$\sum_{v \in V} (d^+(v)f(X_v) + d^-(v)g(X_v)) + \sum_{(i,j) \in A} h(X_i, X_j).$$

The two displayed sums are orthogonal. The h -terms are mutually orthogonal because D has no opposite arcs: disjoint arcs are independent, arcs sharing one endpoint have zero covariance by row or column degeneracy of h . Thus

$$\text{Var} = \sum_{v \in V} \|d^+(v)f + d^-(v)g\|_2^2 + m \|h\|_2^2. \quad (31)$$

The hypothesis makes this variance zero. Hence $h = 0$ and

$$d^+(v)f + d^-(v)g = 0 \quad (32)$$

for every non-isolated v .

We now spell out the elementary linear algebra in (32). If there are two non-isolated vertices v, w such that the vectors

$$(d^+(v), d^-(v)), \quad (d^+(w), d^-(w))$$

are not collinear, then the two equations

$$d^+(v)f + d^-(v)g = 0, \quad d^+(w)f + d^-(w)g = 0$$

form an invertible 2×2 linear system in the two L^2 functions f and g . Hence $f = g = 0$ in L^2 .

It remains to consider the case in which all nonzero degree vectors are collinear. Then there are nonnegative numbers c_v and a fixed vector $(\alpha, \beta) \neq (0, 0)$ such that

$$(d^+(v), d^-(v)) = c_v(\alpha, \beta)$$

for all non-isolated v . Summing over v gives

$$m = \sum_v d^+(v) = \alpha \sum_v c_v, \quad m = \sum_v d^-(v) = \beta \sum_v c_v.$$

Since $m \geq 1$ and some c_v is positive, $\alpha = \beta$. Thus every non-isolated vertex satisfies $d^+(v) = d^-(v)$.

Therefore, if D is not Eulerian, the non-collinear case must occur and $f = g = 0$, so u is constant. If D is Eulerian, then (32) becomes $d(v)(f + g) = 0$ for every non-isolated vertex, with $d(v) > 0$ for at least one such vertex. Hence $f + g = 0$ in L^2 , and therefore $u = c + f(x) - f(y)$. $\square$

5.2 Exact directed rigidity and nonrigidity

Theorem 5.3 (Directed all-arc rigidity with gauge obstruction). *Let $D = (V, A)$ be a finite oriented graph with $m = |A| \geq 1$, and let $W : [0, 1]^2 \rightarrow [0, 1]$ be a directed kernel. Suppose r_a^W is constant for every arc $a \in A$.*

- (i) *If $t(D, W) = 0$, no further conclusion is forced.*
- (ii) *If $t(D, W) > 0$ and D is not Eulerian, then W is constant almost everywhere.*
- (iii) *If $t(D, W) > 0$ and D is Eulerian, then*

$$W(x, y) = p \exp(\phi(x) - \phi(y)) \quad (33)$$

for some $p \in (0, 1]$ and bounded measurable ϕ . Conversely, every kernel of the form (33) with $0 \leq W \leq 1$ has all arc-rooted densities of every Eulerian D equal to the constant p^m .

Proof. Let $Z = t(D, W)$. If $Z = 0$, there is nothing to prove. If $Z > 0$, the same argument as in the graphon case gives $r_a^W \equiv Z$ for every arc and $W \geq Z > 0$. Put $u = \log W$. The Gibbs entropy and Jensen inequality again force

$$\sum_{(i,j) \in A} u(x_i, x_j) = \log Z$$

almost everywhere. Apply Lemma 5.2. This gives either u constant or $u = c + \phi(x) - \phi(y)$. In the Eulerian case put $p = e^c$. Since $u \in L^\infty$, the difference $\phi(x) - \phi(y)$ is essentially bounded. More explicitly, if $|\phi(x) - \phi(y)| \leq M$ for almost every (x, y) , then Fubini gives a point y_0 such that $|\phi(x) - \phi(y_0)| \leq M$ for almost every x . After changing ϕ on a null set and subtracting the constant $\phi(y_0)$, we may take ϕ bounded. Since $W \leq 1$ almost everywhere and the essential supremum of $\phi(x) - \phi(y)$ is nonnegative, this representation has $p \leq 1$. Exponentiating proves the first two assertions and the representation (33).

Conversely, if D is Eulerian and $W(x, y) = p \exp(\phi(x) - \phi(y))$, then for every assignment $\mathbf{x}$,

$$\prod_{(i,j) \in A} W(x_i, x_j) = p^m \prod_{v \in V} \exp((d^+(v) - d^-(v))\phi(x_v)) = p^m.$$

Thus every rooted density is identically p^m . □

Corollary 5.4 (Directed cycles are counterexamples). *Let $C_\ell^\rightarrow$ be a directed cycle. For every nonconstant bounded ϕ and all sufficiently small $p > 0$, the kernel*

$$W(x, y) = p \exp(\phi(x) - \phi(y))$$

is nonconstant, takes values in $[0, 1]$, and has every arc-rooted $C_\ell^\rightarrow$ -density constant.

Proof. A directed cycle is Eulerian. Let

$$M = \text{ess sup } \phi - \text{ess inf } \phi < \infty.$$

If $0 < p \leq e^{-M}$, then $0 \leq p \exp(\phi(x) - \phi(y)) \leq 1$ for almost every (x, y) . Since ϕ is nonconstant, the difference $\phi(x) - \phi(y)$ is nonconstant on a set of positive measure, so W is nonconstant. The converse direction of Theorem 5.3 gives that every arc-rooted density of $C_\ell^\rightarrow$ is identically p^ℓ . □

5.3 Directed stability in the non-Eulerian case

Theorem 5.5 (Directed stability away from the gauge kernel). *Let D be a finite non-Eulerian oriented graph with $m \geq 1$ arcs, and let γ_D be defined by (29). Suppose $0 < \alpha \leq W \leq 1$, put $L = \log(1/\alpha)$, and define, with $q_b^W = r_b^W / t(D, W)$,*

$$\Delta_D(W) = \sum_{b \in A(D)} \|q_b^W - 1\|_{L^1([0,1]^2)}.$$

Then $\gamma_D > 0$ and, with $\bar{u} = \int \log W$,

$$\|\log W - \bar{u}\|_2^2 \leq \max \left\{ \frac{1}{\gamma_D}, \frac{1}{m} \right\} (2e^{mL} + m^2 L^2) L \Delta_D(W). \quad (34)$$

Proof. The positivity $\gamma_D > 0$ follows from the paragraph after (29). Since $W \geq \alpha > 0$, $Z = t(D, W) > 0$ and $u = \log W$ satisfies $-L \leq u \leq 0$. Let

$$S = \sum_{(i,j) \in A(D)} u(X_i, X_j), \quad \bar{u} = \int_{[0,1]^2} u.$$

The D -Gibbs measure has density $Z^{-1}e^S$ with respect to the product Lebesgue measure. Its entropy is

$$D(\mu \parallel \lambda) = \sum_{b \in A(D)} \int_{[0,1]^2} u(x, y) q_b^W(x, y) dx dy - \log Z \geq 0.$$

Put

$$J = \log Z - m\bar{u}.$$

Jensen's inequality gives $J \geq 0$. The entropy inequality gives

$$J \leq \sum_{b \in A(D)} \int_{[0,1]^2} u(x, y) (q_b^W(x, y) - 1) dx dy \leq L \Delta_D(W),$$

because $|u| \leq L$. For $X = S - m\bar{u}$, we have $\mathbb{E}X = 0$, $|X| \leq mL$, and

$$\log \mathbb{E}e^X = \log Z - m\bar{u} = J.$$

By Lemma 2.5,

$$\text{Var}(S) = \text{Var}(X) \leq (2e^{mL} + m^2 L^2) L \Delta_D(W).$$

Use the directed decomposition $u = \bar{u} + f(x) + g(y) + h(x, y)$. Formula (31) and the variational definition (29) yield

$$\text{Var}(S) \geq \gamma_D (\|f\|_2^2 + \|g\|_2^2) + m \|h\|_2^2.$$

Together with Lemma 5.1, this proves (34). $\square$

5.4 Tournamentons

A tournamenton is a measurable function $W : [0, 1]^2 \rightarrow [0, 1]$ such that

$$W(x, y) + W(y, x) = 1 \quad (35)$$

for almost every (x, y) . It is the dense limit analogue of a tournament, i.e. an orientation of a complete graph. If $T = (V, A)$ is a finite tournament, define $t(T, W)$ by the directed formula (26).

Lemma 5.6 (The tournament identity kills the gauge). *Let W be a tournamenton. If*

$$W(x, y) = p \exp(\phi(x) - \phi(y))$$

for some $p > 0$ and measurable ϕ , then $W = 1/2$ almost everywhere.

Proof. The tournament identity gives

$$p \left(e^{\phi(x) - \phi(y)} + e^{\phi(y) - \phi(x)} \right) = 1 \quad (36)$$

for almost every (x, y) . The function $a \mapsto e^a + e^{-a}$ is even and strictly increasing on $[0, \infty)$, so (36) implies that $|\phi(x) - \phi(y)|$ is equal to a constant c for almost every (x, y) . By Fubini, this pairwise relation holds

simultaneously for the three pairs (X, Y) , (Y, Z) , and (Z, X) for almost every triple (X, Y, Z) . If $c > 0$, then the three differences

$$\phi(X) - \phi(Y), \quad \phi(Y) - \phi(Z), \quad \phi(Z) - \phi(X)$$

would each be equal to either c or $-c$, but their sum is identically zero. No sum of three numbers from $\{c, -c\}$ is zero when $c > 0$. Therefore $c = 0$, so ϕ is constant almost everywhere. Then W is constant, and (35) gives $W = 1/2$. $\square$

Theorem 5.7 (All-arc tournamenton rigidity). *Let T be a finite tournament with at least one arc, and let W be a tournamenton. If r_a^W is constant for every arc $a \in A(T)$, then either $t(T, W) = 0$ or $W = 1/2$ almost everywhere.*

Proof. Apply Theorem 5.3 to the oriented graph underlying T . If T is not Eulerian, the directed theorem already gives that W is constant, hence $W = 1/2$ by (35). If T is Eulerian, the directed theorem gives the gauge form, and Lemma 5.6 again gives $W = 1/2$. $\square$

Corollary 5.8 (Arc-transitive tournaments). *Let T be an arc-transitive finite tournament and let $a \in A(T)$. If r_a^W is constant in a tournamenton W , then either $t(T, W) = 0$ or $W = 1/2$ almost everywhere.*

Proof. Every arc of T is the image of a under an automorphism of T , so all arc-rooted densities are constant. Apply Theorem 5.7. $\square$

Corollary 5.9 (The cyclic triangle tournament). *Let $C_3^\rightarrow$ be the cyclic triangle tournament. If*

$$W(x, y) \int_0^1 W(y, z) W(z, x) dz$$

is constant for a tournamenton W , then either $t(C_3^\rightarrow, W) = 0$ or $W = 1/2$ almost everywhere. In the positive case the constant is $1/8$.

Proof. The cyclic triangle is arc-transitive, and the displayed expression is the rooted density of one of its arcs. Apply Corollary 5.8. $\square$

Corollary 5.10 (Stability for nonregular tournaments). *Let T be a non-Eulerian finite tournament with m arcs. Let γ_T be the number γ_D from (29) for the oriented graph underlying T ; by non-Eulerianness, $\gamma_T > 0$. Suppose W is a tournamenton with $0 < \alpha \leq W \leq 1$, put $L = \log(1/\alpha)$, and define, with $q_b^W = r_b^W/t(T, W)$,*

$$\Delta_T(W) = \sum_{b \in A(T)} \|q_b^W - 1\|_{L^1([0,1]^2)}.$$

Then

$$\|W - 1/2\|_2^2 \leq 4 \max\left\{\frac{1}{\gamma_T}, \frac{1}{m}\right\} (2e^{mL} + m^2 L^2) L \Delta_T(W).$$

Proof. Apply Theorem 5.5 to the oriented graph underlying T . Thus, for $u = \log W$ and $\bar{u} = \int u$,

$$\|u - \bar{u}\|_2^2 \leq \max\left\{\frac{1}{\gamma_T}, \frac{1}{m}\right\} (2e^{mL} + m^2 L^2) L \Delta_T(W).$$

Let $p = e^{\bar{u}}$. Since e^x is 1-Lipschitz on $[-L, 0]$,

$$\|W - p\|_2 \leq \|u - \bar{u}\|_2.$$

The tournament identity implies $\int W = 1/2$, so

$$|p - 1/2| = \left| \int (p - W) \right| \leq \|W - p\|_1 \leq \|W - p\|_2.$$

Hence

$$\|W - 1/2\|_2 \leq \|W - p\|_2 + |p - 1/2| \leq 2 \|W - p\|_2.$$

Squaring and using the directed stability bound proves the claim. $\square$

Remark 5.11. Regular tournaments are Eulerian, so the non-Eulerian directed stability theorem does not directly quantify them. Exact rigidity still holds by [Theorem 5.7](#); a sharp stability theorem for regular tournament templates would require a quantitative form of [Lemma 5.6](#). This is a natural problem for future work.

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Declaration on the use of AI

The authors used generative AI tools to assist in discussing proof strategies, checking proofs, and improving exposition. All mathematical arguments, results, and conclusions were reviewed and verified by the authors.

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